UGANDA ADVANCED CERTIFICATE OF EDUCATION

END OF TERM TWO EXAMINATIONS 2025

S.5 BIOLOGY

THEORY

2 hours 30 minutes

Instructions

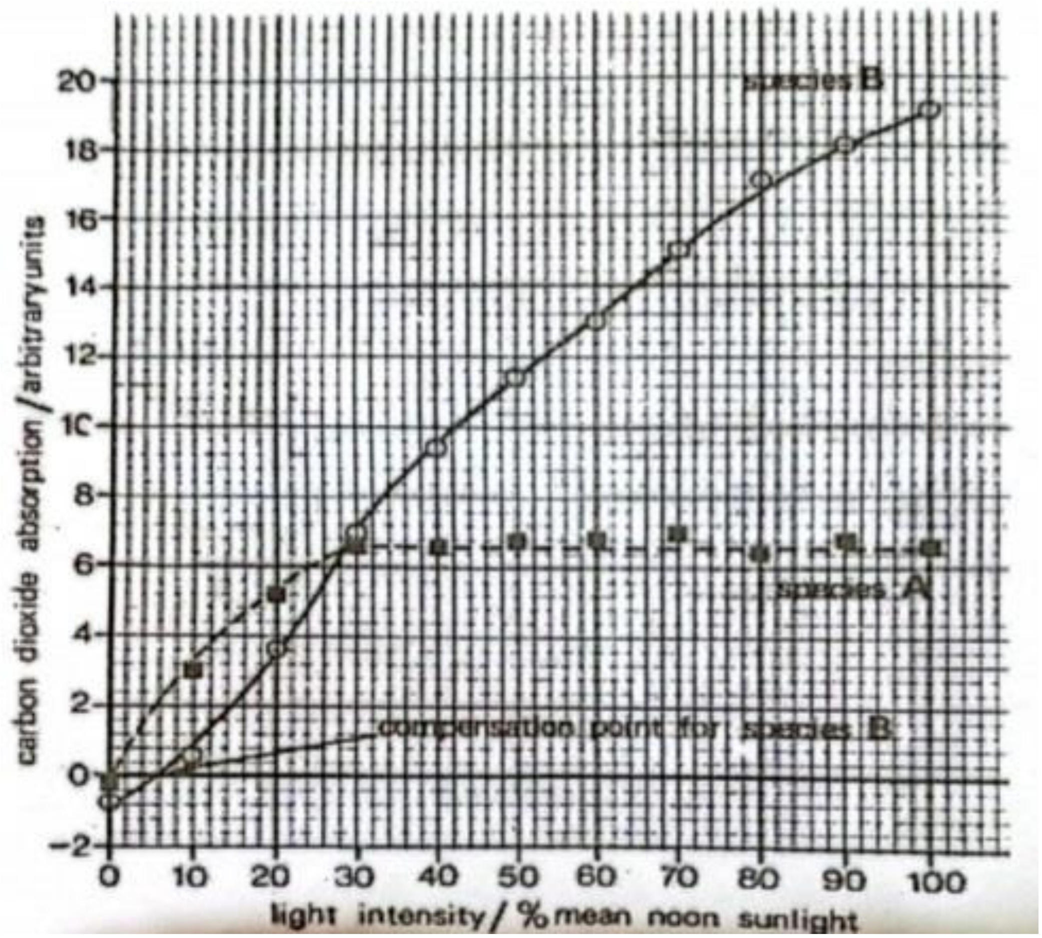
- Attempt all items
- All items carry equal scores
- Use a good handwriting
- Be clear In your answers and direct to the point

For examiners use only

Item	Score	Examiners Comment
1		
2		
3		
4		

Item 1

An investigation was carried out two plant species that grow in two different habitats. With their carbon dioxide absorption rates measured over a range of light intensities. This information is to be used by farmers in south western Uganda and North Eastern Uganda to choose the best crops to grow in their gardens depending on the climate of their areas. South western Uganda has a cool, humid climate with low light intensity while north eastern Uganda receives intense sunlight, low rainfall and is ever hot.
Study the information.



Task

a) Compare the carbon dioxide absorption of the two species of plants used in the investigation.
b) Select the plant species that should be chosen by the farmers in each of the regions, and explain your answer using information shown above.
   i. south western Uganda
   ii. north eastern Uganda

c) What are the biological identity of species A and B used in the investigation and for each give examples of crops that show such kinds of physiological behavior.
d) What physiological adaptations does species B have to survive in the stated environment?

Item two

A group of hikers is stranded on a high-altitude mountain (4,500 meters above sea level). One of the hikers, Sarah, is experiencing severe sickness—she is dizzy, short of breath, and confused. The rescue team needs to stabilize her before evacuating her to a lower altitude.

Task

a) Explain why Sarah is experiencing the above symptoms
b) Suggest possible measures the rescue team can use to stabilize her before evacuation to low altitude. Explain your answers using biological reasoning.
c) If she they remain stranded on the mountain for weeks, explain how Sarah's body might adjust to make easily survive in that condition.
d) How can the problems faced by Sarah be avoided in future while climbing very high mountains.

Item 3

Two babies Cathy and Leonard are each 5 years old. Cathy was immunized of the common diseases as scheduled by the ministry of health, while Leonard was not immunized because his parents hesitated to do so.
Recently, there was a measles outbreak and both children contracted the illness, Cathy had mild fever and rash while Leonard had severe symptoms of fever, cough, conjunctivitis and intense rash and he nearly died.

Task

a) Describe the immunological processes that occurred in Cathy's body from the time she was immunized for measles up to the time she overcame the infection during the outbreak.
b) Explain why Leonard experienced a more severe attack than Cathy.
c) There are many different strains of the rhinovirus, which causes the common cold. Explain why people can catch several different colds in the space of a few months.

Item 4

A small dairy farm has been producing fresh milk for years, but suddenly, their milk is spoiling much faster than usual—sometimes within a day, even when refrigerated. The farmers suspect that something is interfering with the natural enzyme activity that helps preserve milk, such as contamination with bacteria and salts that remain on the milking machines. The Bacteria reproduce faster and survive in milk even when it is of different concentrations

Task

a) Explain how bacterial contamination and salt remains on utensils can result into faster milk spoilage.
b) Which alternative methods can the farm use so as to prevent spoilage of their milk in the future?
c) What adaptations do bacteria have to easily survive in the milk and cause it's spoilage?

END